

Young Tableaux

Young tableaux are combinatorial pictures of numbered boxes, and for a physicist they are the one bookkeeping device that links three things you already use: the symmetry types of tensors with n indices, the irreducible blocks of n -qubit Hilbert space under collective rotations, and the explicit projectors that pull each symmetry sector out of a state. This note builds the picture from scratch using the paclet's `Wolfram`TensorNetworks`Symmetry`` subcontext, with quantum-state examples (Bell, singlet, $|W\rangle$, $|001\rangle$) constructed through the TN paclet.

Why a Young diagram?

A Young diagram is a picture of an integer partition: boxes arranged in left-justified rows of weakly decreasing length. Why a quantum researcher should care, in one picture.

Take n qubits and consider the operations that act on all of them at once: a global $SU(2)$ rotation $U^{\otimes n}$ (the same rotation on each qubit), and a permutation that shuffles the n qubits. These two families of operations commute with each other, and together they split the Hilbert space into independent blocks that neither family can mix. The block labels are the Young diagrams of size n . Each diagram pins down one specific total spin J , and within that spin sector there may be several independent copies (one per standard filling of the diagram).

Anything in your physics problem that has this collective symmetry (a Hamiltonian invariant under global $SU(2)$, noise that hits all qubits identically, n identical copies of one quantum state) *must* respect the decomposition. So before solving anything, you can replace "diagonalize a $2^n \times 2^n$ matrix" with "solve one much smaller problem inside each Young-diagram block". That is the only reason for the rest of this note.

Four concrete payoffs land in later sections, each one phrased here in language you already use.

- 1. Decoherence-free subspaces (DFS).** If the noise on n qubits is collective (the same single-qubit unitary V on every qubit as $V^{\otimes n}$), there are subspaces of (that the noise leaves pointwise fixed. A logical qubit encoded in such a subspace is automatically protected against collective dephasing, with no syndrome extraction. Young diagrams enumerate exactly those protected subspaces.
- 2. Spectrum estimation from independent and identically distributed samples.** Given n identical copies of an unknown single-qubit state ρ , measuring in a particular basis returns one discrete outcome (a Young diagram) whose distribution concentrates on $\text{spec}(\rho)$ as $n \rightarrow \infty$. The Young diagram is the *universal sufficient statistic* for spectrum estimation from independent and identically distributed samples, and the outcome probability for any given state is a single projection norm.

3. *Symmetric tensor networks*. An MPS / PEPS / MERA ansatz that preserves SU(2) (or any compact symmetry) has bond Hilbert spaces that automatically split into Young-diagram blocks. Contractions become block-diagonal, and the ansatz produces states with the correct total quantum-number sector by construction. The per-block sizes are computable *before* you build any tensor.
4. *Total angular momentum*. For n qubits, the Hilbert space (decomposes into total- J multiplets, the rank- n analog of two-spin Clebsch-Gordan addition. Which spin- J appears, how many times, and with what block size, all read off Young diagrams.

These four are not separate tricks. They are *one* construction (replace an n -qubit problem with one independent problem per Young-diagram block) read with four different physical questions in mind. The rest of this note builds the machinery in the order each payoff needs it: partitions and diagrams (the language), hook-length and hook-content formulas (the block sizes), projection operators built from a single tableau (how to land a state in one block), and worked quantum-state examples that realize each of the four payoffs above.

Load the paclet and pull the Symmetry subcontext into scope; both Needs are assumed in every subsequent cell.

```
In[1]:= PacletDirectoryLoad["Users/mohammadb/Documents/GitHub/TensorNetworks"];
Needs["Wolfram`TensorNetworks`"];
Needs["Wolfram`TensorNetworks`Symmetry`"];
```

Integer partitions

A *partition* of a positive integer n is a weakly decreasing tuple $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_k\}$ of positive integers summing to n ; we write $\lambda \vdash n$. The Wolfram Language enumerates them with `IntegerPartitions[n]`.

```
In[4]:= IntegerPartitions[4]
Out[4]= {{4}, {3, 1}, {2, 2}, {2, 1, 1}, {1, 1, 1, 1}}
```

There are five partitions of $n = 4$:

$\{\{4\}, \{3, 1\}, \{2, 2\}, \{2, 1, 1\}, \{1, 1, 1, 1\}\}$. Each one will become a Young diagram, and each diagram labels one irreducible representation of S_4 .

The paclet provides a predicate `PartitionQ` that decides whether a list of integers is a valid partition (positive entries, weakly decreasing). Pick $\{3, 1\}$ from `IntegerPartitions[4]`: it passes.

```
In[5]:= PartitionQ[{3, 1}]
Out[5]= True
```

Swap the same entries into $\{1, 3\}$ and the weakly-decreasing condition fails.

```
In[6]:= PartitionQ[{1, 3}]
```

```
Out[6]= False
```

Drawing the diagram

The diagram of $\lambda = \{\lambda_1, \dots, \lambda_k\}$ is drawn (in English notation) as k left-justified rows with λ_i boxes in row i . The packet's typed handle is `YoungTableau`: in *partition form*, `YoungTableau[lambda]` fills the boxes with slot labels 1, 2, ..., n row by row and renders the shape. Pick a few extreme shapes to build intuition.

A single row $\{n\}$ is the *fully symmetric* shape on n slots.

```
In[7]:= YoungTableau[{4}]
```

```
Out[7]= YoungTableau[
  + 1 2 3 4 Shape: {4}
      Filling: Canonical
      Dimension: 1
]
```

A single column $\{1, 1, \dots, 1\}$ (n ones) is the *fully antisymmetric* shape.

```
In[8]:= YoungTableau[{1, 1, 1, 1}]
```

```
Out[8]= YoungTableau[
  + 1 Shape: {1, 1, 1, 1}
    2 Filling: Canonical
    3 Dimension: 1
    4
]
```

A square shape $\{2, 2\}$ has rank 4 and is neither symmetric nor antisymmetric: it is "mixed".

```
In[9]:= YoungTableau[{2, 2}]
```

```
Out[9]= YoungTableau[
  + 1 2 Shape: {2, 2}
    3 4 Filling: Canonical
      Dimension: 2
]
```

A staircase $\{3, 2, 1\}$ has rank 6 and is the densest non-rectangular shape at its size.

```
In[10]:= YoungTableau[{3, 2, 1}]
```

```
Out[10]= YoungTableau[
  + 1 2 3 Shape: {3, 2, 1}
    4 5 Filling: Canonical
      6 Dimension: 16
]
```

Shape, size, and the column reading

Four inspectors read off the structure of a `YoungTableau`. `TableauShape` returns the underlying partition.

```
In[11]:= TableauShape[YoungTableau[{3, 2, 1}]]
Out[11]= {3, 2, 1}
```

TableauSize returns the total box count, which is also the tensor rank the tableau can act on.

```
In[12]:= TableauSize[YoungTableau[{3, 2, 1}]]
Out[12]= 6
```

TableauColumns is the one that earns its keep on ragged shapes: it returns the slot indices column by column, correctly stopping each column when the diagram does. Compare row reading and column reading for the L-shape {4, 2, 1}.

```
In[13]:= TableauRows[YoungTableau[{4, 2, 1}]]
Out[13]= {{1, 2, 3, 4}, {5, 6}, {7}}
```

And the column reading on the same shape (ragged: column 4 stops after row 1).

```
In[14]:= TableauColumns[YoungTableau[{4, 2, 1}]]
Out[14]= {{1, 5, 7}, {2, 6}, {3}, {4}}
```

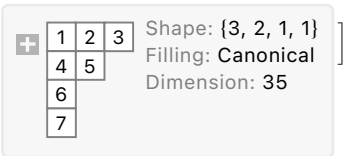
Conjugate (transpose) diagram

Swapping rows and columns of a Young diagram gives its *conjugate* (or *transpose*) λ^T : the box at $\{i, j\}$ moves to $\{j, i\}$, and TransposePartition returns the new partition.

```
In[15]:= TransposePartition[{4, 2, 1}]
Out[15]= {3, 2, 1, 1}
```

Visually: {4, 2, 1} has rows of lengths 4, 2, 1, while its conjugate {3, 2, 1, 1} has rows of lengths 3, 2, 1, 1.

```
In[16]:= YoungTableau[TransposePartition[{4, 2, 1}]]
Out[16]=
```

YoungTableau [ Shape: {3, 2, 1, 1}
Filling: Canonical
Dimension: 35]

Conjugation is an involution, $\lambda^{TT} = \lambda$, and it exchanges "symmetric in slot i " with "antisymmetric in slot i " wherever it appears, which is why $\{n\}$ and $\{1, 1, \dots, 1\}$ are conjugates: fully symmetric and fully antisymmetric on n slots.

```
In[17]:= TransposePartition[TransposePartition[{4, 2, 1}]]
Out[17]= {4, 2, 1}
```

Standard Young tableaux

A *standard Young tableau* (SYT) of shape $\lambda \vdash n$ is a filling of the n boxes with $\{1, 2, \dots, n\}$ (each used exactly once) that strictly increases left-to-right along every row *and* top-to-bottom down every column. The packet's *explicit-rows form* `YoungTableau[{rows}]` accepts any such filling and exposes it for downstream projectors.

```
In[18]:= YoungTableau[{{1, 2, 3}, {4, 5}}]
Out[18]=
```

+

1	2	3
4	5	

Shape: {3, 2}
 Filling: Canonical
 Dimension: 5

A different SYT of the *same* shape is a different filling.

```
In[19]:= YoungTableau[{{1, 2, 4}, {3, 5}}]
Out[19]=
```

+

1	2	4
3	5	

Shape: {3, 2}
 Filling: Standard
 Dimension: 5

The "standard" qualifier earns its keep: not every filling of a valid diagram with the labels $1, \dots, n$ counts. Here the first row decreases, so the strict-increase rule fails along that row.

```
In[20]:= YoungTableau[{{3, 2, 1}, {4, 5}}]
Out[20]=
```

+

3	2	1
4	5	

Shape: {3, 2}
 Filling: Explicit
 Dimension: 5

Here the columns are the problem: column 1 reads 4 above 1, which decreases top-to-bottom.

```
In[21]:= YoungTableau[{{4, 2, 3}, {1, 5}}]
Out[21]=
```

+

4	2	3
1	5	

Shape: {3, 2}
 Filling: Explicit
 Dimension: 5

A completely shuffled filling fails on both rows and columns at once.

```
In[22]:= YoungTableau[{{5, 1, 3}, {2, 4}}]
Out[22]=
```

+

5	1	3
2	4	

Shape: {3, 2}
 Filling: Explicit
 Dimension: 5

The predicate `YoungTableauQ` is *structural*: it checks that the shape is a valid partition and that the entries are exactly $\{1, \dots, n\}$ used once each, but it does not enforce the strict-increase rule. All four fillings above pass.

```
In[23]:= YoungTableauQ /@ {YoungTableau[{{1, 2, 4}, {3, 5}}],
  YoungTableau[{{3, 2, 1}, {4, 5}}],
  YoungTableau[{{4, 2, 3}, {1, 5}}],
  YoungTableau[{{5, 1, 3}, {2, 4}}]}
```

```
Out[23]= {True, True, True, True}
```

It does reject fillings that violate the shape (here, the bottom row is longer than the top).

```
In[24]:= YoungTableauQ[{{1}, {2, 3}}]
Out[24]= False
```

The paclet exposes a separate predicate `StandardTableauQ` that enforces the strict-increase rule (and requires `YoungTableauQ` to pass first). Apply it to the same four fillings: only the first is a true SYT.

```
In[25]:= AssociationMap[StandardTableauQ, YoungTableau /@ {{{1, 2, 4}, {3, 5}},
  {{3, 2, 1}, {4, 5}},
  {{4, 2, 3}, {1, 5}},
  {{5, 1, 3}, {2, 4}}}]
```

```
Out[25]= {
  YoungTableau[
    + 

|   |   |   |
|---|---|---|
| 1 | 2 | 4 |
| 3 | 5 |   |


    Shape: {3, 2}
    Filling: Standard
    Dimension: 5
  ] → True,
  YoungTableau[
    + 

|   |   |   |
|---|---|---|
| 3 | 2 | 1 |
| 4 | 5 |   |


    Shape: {3, 2}
    Filling: Explicit
    Dimension: 5
  ] → False,
  YoungTableau[
    + 

|   |   |   |
|---|---|---|
| 4 | 2 | 3 |
| 1 | 5 |   |


    Shape: {3, 2}
    Filling: Explicit
    Dimension: 5
  ] → False,
  YoungTableau[
    + 

|   |   |   |
|---|---|---|
| 5 | 1 | 3 |
| 2 | 4 |   |


    Shape: {3, 2}
    Filling: Explicit
    Dimension: 5
  ] → False
}
```

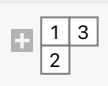
The count of standard fillings of shape λ equals $\dim V_\lambda$ and is what `TableauDimension[lambda]` returns.

Two SYT, two copies: the physics reading

Why care about distinct fillings of the same shape? Because they label distinct copies of the same irreducible representation inside $V^{\otimes n}$. For three qubits the shape $\{2, 1\}$ has exactly two SYTs, and that count is the multiplicity with which $\text{spin-}\frac{1}{2}$ appears in (under collective $\text{SU}(2)$).

```
In[26]:= {YoungTableau[{{1, 2}, {3}}], YoungTableau[{{1, 3}, {2}}]}
Out[26]=
```


 Shape: {2, 1}
 Filling: Canonical
 Dimension: 2


 Shape: {2, 1}
 Filling: Standard
 Dimension: 2

The same number falls out of the hook length formula, computed next.

```
In[27]:= TableauDimension[{2, 1}]
Out[27]=
2
```

This multiplicity is not bookkeeping for its own sake: it is the dimension of the protected subspace a quantum-information protocol can use. The DFS section below shows multiplicity-2 giving a noise-immune logical qubit and the Schur-sampling section below shows the same multiplicity space being the freedom that spectrum-estimation protocols deliberately discard.

Counting SYT: the hook length formula

The *hook* of a box $\{i, j\} \in \lambda$ collects the box itself, the boxes to its right in the same row, and the boxes below it in the same column; its size is the *hook length* $h_\lambda(i, j) = (\lambda_i - j) + (\lambda_j^T - i) + 1$. Read off all hook lengths of a small shape.

```
In[28]:= HookLengths[{3, 2}]
Out[28]=
{{4, 3, 1}, {2, 1}}
```

Read a single hook length.

```
In[29]:= HookLength[YoungTableau[{3, 2}], {1, 1}]
Out[29]=
4
```

The Frame-Robinson-Thrall formula collapses the irreducible-representation dimension into

$$f^\lambda = \dim V_\lambda = \frac{n!}{\prod_{\{i,j\} \in \lambda} h_\lambda(i, j)},$$

evaluated by the paclet via the Frobenius determinant ($O(r^3)$ in the row count, much faster than naive). `HookFactor` returns the reciprocal hook product and `TableauDimension` multiplies by $n!$.

```
In[30]:= HookFactor[{3, 2}]
Out[30]=

$$\frac{1}{24}$$

```

And `TableauDimension` multiplies by $n!$ to give the full irrep dimension f^λ .

```
In[31]:= TableauDimension[{3, 2}]
Out[31]=
5
```

SchurDimension: the $GL(d)$ block size

A *semistandard Young tableau* (SSYT) of shape λ with entries in $\{1, \dots, d\}$ has weakly increasing rows and strictly increasing columns; its count equals $\dim W_\lambda(d)$, the dimension of the irreducible polynomial $GL(d)$ -representation W_λ . The closed form is Stanley's *hook-content* formula

$$\dim W_\lambda(d) = \prod_{\{i,j\} \in \lambda} \frac{d+j-i}{h_\lambda(i,j)},$$

returned by `SchurDimension[lambda, d]`. For qubits ($d = 2$), $W_\lambda(2)$ is the spin- j representation with $j = (\lambda_1 - \lambda_2)/2$.

The fully symmetric shape $\{3\}$ on three qubits gives the spin- $\frac{3}{2}$ representation (dimension 4).

```
In[32]:= SchurDimension[{3}, 2]
Out[32]=
4
```

The mixed shape $\{2, 1\}$ on three qubits gives the spin- $\frac{1}{2}$ representation (dimension 2).

```
In[33]:= SchurDimension[{2, 1}, 2]
Out[33]=
2
```

The fully antisymmetric shape $\{1, 1, 1\}$ vanishes on qubits because three antisymmetric slots cannot all carry distinct values from $\{0, 1\}$.

```
In[34]:= SchurDimension[{1, 1, 1}, 2]
Out[34]=
0
```

At $d = 3$ (qutrits), the same shape is one-dimensional (the unique antisymmetric tensor ϵ_{ijk}).

```
In[35]:= SchurDimension[{1, 1, 1}, 3]
Out[35]=
1
```


Schur-Weyl bookkeeping on n qubits

Schur-Weyl duality decomposes $V^{\otimes n}$ as a $GL(d) \times S_n$ bimodule,

$$V^{\otimes n} \cong \bigoplus_{\lambda \vdash n} W_\lambda \otimes V_\lambda,$$

$\ell(\lambda) \leq d$

so dimensions satisfy $d^n = \sum_\lambda \dim W_\lambda(d) \cdot \dim V_\lambda$. For n qubits this reads as "spin- j appears with multiplicity $\dim V_\lambda$ ". The next three cells expand the sum for $n = 2, 3, 4$ qubits.

```
In[36]:= With[{parts = Select[IntegerPartitions[2], Length[#] ≤ 2 &]},
  Grid[Prepend[
    {#, TableauDimension[#], SchurDimension[#], 2],
    TableauDimension[#] × SchurDimension[#], 2]} & /@ parts,
  {"λ", Row[{"dim ", Subscript["V", "λ"]]},
  Row[{"dim ", Subscript["W", "λ"], "(2)"}],
  Row[{"dim ", Subscript["V", "λ"], " · dim ",
    Subscript["W", "λ"], "(2)"}]], Frame → All]]
```

Out[36]=

λ	$\dim V_\lambda$	$\dim W_\lambda(2)$	$\dim V_\lambda \cdot \dim W_\lambda(2)$
$\{2\}$	1	3	3
$\{1, 1\}$	1	1	1

Two qubits split as $2^2 = 3 + 1$: one copy of spin-1 (the triplet, $\dim W = 3$, from shape $\{2\}$) plus one copy of spin-0 (the singlet, $\dim W = 1$, from shape $\{1, 1\}$). Both shapes have multiplicity 1 because each has exactly one SYT.

Now $n = 3$ qubits.

```
In[37]:= With[{parts = Select[IntegerPartitions[3], Length[#] ≤ 2 &]},
  Grid[Prepend[
    {#, TableauDimension[#], SchurDimension[#], 2],
    TableauDimension[#] × SchurDimension[#], 2]} & /@ parts,
  {"λ", Row[{"dim ", Subscript["V", "λ"]]},
  Row[{"dim ", Subscript["W", "λ"], "(2)"}],
  Row[{"dim ", Subscript["V", "λ"], " · dim ",
    Subscript["W", "λ"], "(2)"}]], Frame → All]]
```

Out[37]=

λ	$\dim V_\lambda$	$\dim W_\lambda(2)$	$\dim V_\lambda \cdot \dim W_\lambda(2)$
$\{3\}$	1	4	4
$\{2, 1\}$	2	2	4

Three qubits split as $2^3 = 4 + 4 = 8$: one copy of spin- $\frac{3}{2}$ (the quadruplet, $\dim W = 4$, from $\{3\}$) plus *two* copies of spin- $\frac{1}{2}$ ($\dim W = 2$, from $\{2, 1\}$, which has two SYTs). The mult column 2 for shape $\{2, 1\}$ is exactly the number of distinct standard fillings encountered earlier.

And $n = 4$ qubits.

```
In[38]:= With[{parts = Select[IntegerPartitions[4], Length[#] ≤ 2 &]},
  Grid[Prepend[
    {#, TableauDimension[#], SchurDimension[#], 2},
    {TableauDimension[#] × SchurDimension[#], 2} & /@ parts,
    {"λ", Row[{"dim ", Subscript["V", "λ"]}],
    Row[{"dim ", Subscript["W", "λ"], "(2)"}],
    Row[{"dim ", Subscript["V", "λ"], " · dim ",
    Subscript["W", "λ"], "(2)"}]]], Frame → All]]
```

Out[38]=

λ	$\dim V_\lambda$	$\dim W_\lambda(2)$	$\dim V_\lambda \cdot \dim W_\lambda(2)$
{4}	1	5	5
{3, 1}	3	3	9
{2, 2}	2	1	2

Four qubits split as $2^4 = 5 + 9 + 2 = 16$: one copy of spin-2 ($\dim W = 5$, from {4}), *three* copies of spin-1 ($\dim W = 3$, from {3, 1} which has three SYTs), and *two* copies of spin-0 ($\dim W = 1$, from {2, 2} which has two SYTs). In every row, the product column is $\dim V_\lambda \cdot \dim W_\lambda(2) = (\text{multiplicity}) \times (\text{spin} - j \text{ block size})$, and the column sums to 2^n .

Young projectors

The *Young symmetrizer* for a tableau T of shape λ is the group-algebra element $c_T = a_T b_T \in$, with a_T summing over the row stabilizer and b_T over the column stabilizer with signs; it is idempotent up to scale, $c_T^2 = (n! / f^\lambda) c_T$, and the normalized projector

$$P_T = \frac{f^\lambda}{n!} c_T, \quad P_T^2 = P_T,$$

lands a tensor in one specific copy of the λ -irrep inside $V^{\otimes n}$. The paclet exposes

`YoungSymmetrize[T, tab] (c_T · T)` and `YoungProject[T, tab] (P_T · T)`.

Two operational facts used below. (1) A *single-tableau* P_T is an algebraic idempotent but *not* an orthogonal Hermitian projector, so single-tableau norms do not satisfy Pythagoras. (2) Summing those single-tableau projectors over the standard tableaux, $M = \sum_{T \in \text{SYT}(\lambda)} P_T$ gives an operator whose range is the entire λ -block but which is itself generally not idempotent; the orthogonal projector onto that block is the genuine Hermitian idempotent $P_\lambda = M M^+ (M^+ \text{ the Moore-Penrose pseudoinverse})$. For n qubits the shape $\lambda = (\lambda_1, \lambda_2)$ labels the total- J sector with $J = (\lambda_1 - \lambda_2)/2$, so $\|P_\lambda | \psi\rangle\|^2$ is the probability of measuring total spin J in state $| \psi\rangle$. Define a helper for that probability.

```
In[39]:= isotypicProjector[syts_List] :=
  isotypicProjector[syts] = Module[{n = TableauSize[First[syts]], m},
    m = (Function[ket, Flatten[Total[(YoungProject[ket, #1] &) /@ syts]]] /@
      (Normal[SparseArray[#1 → 1, ConstantArray[2, n]]] &) /@ Tuples[{1, 2}, n])T
    m = If[n ≤ 5, m, N[m]];
    m.m+]
```

```
In[40]:= isoNormSq[psi_, syts_List] :=
  With[{p = isotypicProjector[syts]}, Re[Conjugate[Flatten[psi]].p.Flatten[psi]]]
```

In closed form, `isoNormSq[psi, syts]` returns

$$\|P_\lambda |\psi\rangle\|^2 = \langle \psi | P_\lambda | \psi \rangle$$

(equal because P_λ is a Hermitian idempotent), with the orthogonal isotypic projector

$$P_\lambda = M M^+, \quad M = \sum_T P_T$$

summed over $T \in SYT(\lambda)$, and each single-tableau Young projector

$$P_T = \frac{f^\lambda}{n!} c_T, \quad c_T = a_T b_T = \left(\sum_{\sigma \in R_T} \sigma \right) \left(\sum_{\tau \in C_T} sgn(\tau) \tau \right).$$

Here R_T and C_T are the row and column stabilizers of T , $f^\lambda = \dim V_\lambda$ (= number of SYTs of shape $\lambda = \text{TableauDimension}[\lambda]$), and each $\sigma \in S_n$ acts on $V^{\otimes n}$ by permuting the n tensor slots. The isotypic projector P_λ has the equivalent character form (a general identity for finite groups),

$$P_\lambda = \frac{f^\lambda}{n!} \sum_{\sigma \in S_n} \chi_\lambda(\sigma) \sigma,$$

where χ_λ is the irreducible S_n -character. Unlike the individual P_T (idempotent but not self-adjoint), P_λ is a genuine orthogonal Hermitian projector, so

$$\|P_\lambda |\psi\rangle\|^2 = \text{Pr}(S_{tot}^2 = J(J+1) \mid |\psi\rangle)$$

with $J = (\lambda_1 - \lambda_2)/2$.

Worked example. For $|\text{GHZ}_3\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$ and shape $\{3\}$ with the unique SYT

$T = \{\{1, 2, 3\}\}$: $R_T = S_3$ and $C_T = \{e\}$ (every column has length 1), so $c_T = \sum_{\sigma \in S_3} \sigma$ and $P_{\{3\}} = \frac{1}{6} \sum_{\sigma} \sigma$ is the full symmetrizer. Each σ leaves $|\text{GHZ}_3\rangle$ invariant, so $P_{\{3\}} |\text{GHZ}_3\rangle = |\text{GHZ}_3\rangle$ and

$\|P_{\{3\}} |\text{GHZ}_3\rangle\|^2 = 1$. For shape $\{2, 1\}$ with $T_A = \{\{1, 2\}, \{3\}\}$, $c_{T_A} = (e + (12))(e - (13))$; each computational state $|i i i\rangle$ is invariant under both (12) and (13), so $(e - (13)) |\text{GHZ}_3\rangle = 0$ and $c_{T_A} |\text{GHZ}_3\rangle = 0$.

The same kills c_{T_B} , giving $P_{\{2,1\}} |\text{GHZ}_3\rangle = 0$. These two algebraic facts ($\|P_{\{3\}} |\text{GHZ}_3\rangle\|^2 = 1$,

$\|P_{\{2,1\}} |\text{GHZ}_3\rangle\|^2 = 0$) are exactly what the GHZ₃ cells below produce.

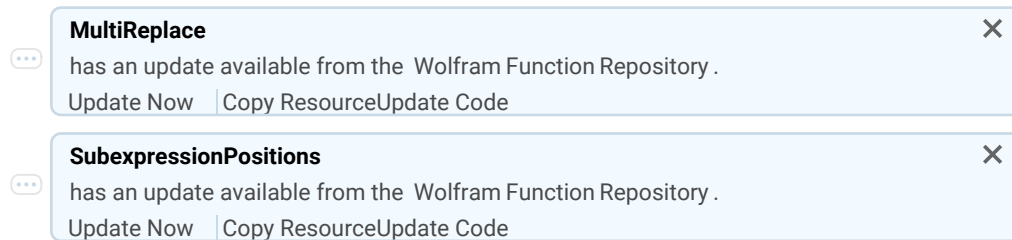
GHZ₃ and W₃: same total- J , different M

The 3-qubit GHZ state is a *genuine* tensor-network node: a COPY/delta tensor where three legs share one bond, so contracting that bond forces all three physical indices equal. Build it that way through the packet (three rank-2 tensors hyperedged on bond "a", normalized so the contraction is $\delta_{ijk} / \sqrt{2} = (|000\rangle + |111\rangle) / \sqrt{2}$).

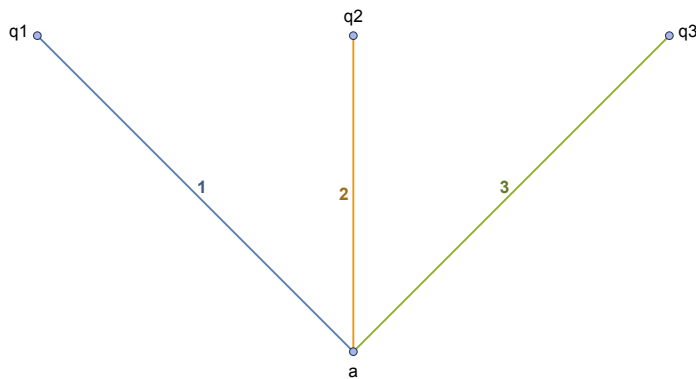
```
In[41]:= ghz3TN = TensorNetwork[
  {IdentityMatrix[2] / 2^(1/6), IdentityMatrix[2] / 2^(1/6),
   IdentityMatrix[2] / 2^(1/6)},
  {{{"q1", "a"}, {"q2", "a"}, {"q3", "a"}},
   {"q1", "q2", "q3"}}];
```

Inspect the hypergraph: three rank-2 tensors all bonded on the shared edge "a", with three free physical legs.

```
In[42]:= ghz3TN["Hypergraph"]
```



Out[42]=



Contract to the state array.

```
In[43]:= ghz3 = TensorNetworkContract @ ghz3TN
```

Out[43]=

$$\left\{ \left\{ \left\{ \frac{1}{\sqrt{2}}, 0 \right\}, \{0, 0\} \right\}, \left\{ \{0, 0\}, \left\{ 0, \frac{1}{\sqrt{2}} \right\} \right\} \right\}$$

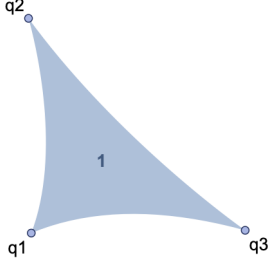
The two nonzero amplitudes $|000\rangle$ and $|111\rangle$ identify this as the superposition $|J = \frac{3}{2}, M = \pm \frac{3}{2}\rangle$.

The W state has three nonzero amplitudes at the one-excitation strings; load it as a single rank-3 tensor through the packet.

```
In[44]:= w3TN = TensorNetwork[
  {Normal @
    SparseArray[{{1, 1, 2}, {1, 2, 1}, {2, 1, 1}} → 1/Sqrt[3], {2, 2, 2}]],
  {"q1", "q2", "q3"}], {"q1", "q2", "q3"}];
```

The hypergraph is a single rank-3 node with three free legs (no internal bonds; W has no nontrivial tensor-network factorization).

```
In[45]:= w3TN["Hypergraph"]
Out[45]=
```



Contract.

```
In[46]:= w3 = TensorNetworkContract @ w3TN
Out[46]=
```

$$\left\{ \left\{ \left\{ 0, \frac{1}{\sqrt{3}} \right\}, \left\{ \frac{1}{\sqrt{3}}, 0 \right\} \right\}, \left\{ \left\{ \frac{1}{\sqrt{3}}, 0 \right\}, \{0, 0\} \right\} \right\}$$

The equal-amplitude sum over single-flip strings is exactly $|J = \frac{3}{2}, M = \frac{1}{2}\rangle$.

Both states are fully Bose-symmetric, so projecting onto the unique SYT of shape $\{3\}$ returns each one unchanged. GHZ first.

```
In[47]:= YoungProject[ghz3, YoungTableau[{{1, 2, 3}}]]
Out[47]=
```

$$\left\{ \left\{ \left\{ \frac{1}{\sqrt{2}}, 0 \right\}, \{0, 0\} \right\}, \left\{ \{0, 0\}, \left\{ 0, \frac{1}{\sqrt{2}} \right\} \right\} \right\}$$

Then W.

```
In[48]:= YoungProject[w3, YoungTableau[{{1, 2, 3}}]]
Out[48]=
```

$$\left\{ \left\{ \left\{ 0, \frac{1}{\sqrt{3}} \right\}, \left\{ \frac{1}{\sqrt{3}}, 0 \right\} \right\}, \left\{ \left\{ \frac{1}{\sqrt{3}}, 0 \right\}, \{0, 0\} \right\} \right\}$$

Young projectors see the *shape* λ (equivalently, the total J), not the magnetization M , so both states being fixed by $P_{\{3\}}$ confirms they live in the spin- $\frac{3}{2}$ block.

Projecting either state onto any SYT of shape $\{2, 1\}$ gives zero: there is no spin- $\frac{1}{2}$ content. GHZ:

```
In[49]:= YoungProject[ghz3, YoungTableau[{{1, 2}, {3}}]] // Flatten // Norm
Out[49]=
```

0

And W:

```
In[50]:= YoungProject[w3, YoungTableau[{{1, 2}, {3}}]] // Flatten // Norm
Out[50]=
0
```

GHZ and W therefore both live entirely in the spin- $\frac{3}{2}$ block. Young projection cannot separate them; to distinguish W from GHZ one needs an *additional* operator inside that block (e.g., S_z^{tot}).

A product state $|+\rangle \otimes |0\rangle \otimes |+\rangle$: nontrivial J -spectrum

A product state has zero entanglement, yet $\vec{S}_{\text{tot}}^2$ does not have a definite eigenvalue on it: adding three spin- $\frac{1}{2}$ s quantum-mechanically gives $J = \frac{3}{2}$ and $J = \frac{1}{2}$ pieces both. Build $|+\rangle \otimes |0\rangle \otimes |+\rangle$ as a TN of three rank-1 single-qubit tensors with no internal bonds.

```
In[51]:= prodPlus0PlusTN = TensorNetwork[
  {{1, 1} / Sqrt[2], {1, 0}, {1, 1} / Sqrt[2]},
  {"q1"}, {"q2"}, {"q3"}],
  {"q1", "q2", "q3"}];
```

Three disconnected rank-1 tensors (no internal bonds), one free leg each.

```
In[52]:= prodPlus0PlusTN["Hypergraph"]
Out[52]=
```



Contract.

```
In[53]:= prodPlus0Plus = TensorNetworkContract @ prodPlus0PlusTN
Out[53]=
```

```
{{{1/2, 1/2}, {0, 0}}, {{1/2, 1/2}, {0, 0}}}
```

Spin- $\frac{3}{2}$ (shape {3}) probability:

```
In[54]:= isoNormSq[prodPlus0Plus, {YoungTableau[{{1, 2, 3}}]}]
Out[54]=
2/3
```

Spin- $\frac{1}{2}$ (shape {2, 1}, summed over its two SYTs) probability:

```
In[55]:= isoNormSq[prodPlus0Plus,
  {YoungTableau[{{1, 2}, {3}}], YoungTableau[{{1, 3}, {2}}]}]
Out[55]=

$$\frac{1}{3}$$

```

Physics readout: the two numbers are $2/3$ and $1/3$, and they sum to 1. Analytical check:

$|+\rangle \otimes |0\rangle \otimes |+\rangle = \frac{1}{2} (|000\rangle + |001\rangle + |100\rangle + |101\rangle)$, so its overlap with the W-like state $|J = \frac{3}{2}, M = \frac{1}{2}\rangle = (|001\rangle + |010\rangle + |100\rangle) / \sqrt{3}$ is $(\frac{1}{2} + 0 + \frac{1}{2}) / \sqrt{3} = 1 / \sqrt{3}$, contributing $1/3$ to $|c_{3/2}|^2$; summing over all M in the spin- $\frac{3}{2}$ block recovers $2/3$. A *product* state populates both available total- J sectors, and Young projection delivers exactly the Clebsch-Gordan amplitudes.

A 4-qubit basis state $|1100\rangle$: three J sectors at $M = 0$

The $M = 0$ subspace of 4 qubits has dimension $\binom{4}{2} = 6$ and decomposes by Schur-Weyl as $J = 2$ (one state at $M = 0$), $J = 1$ (multiplicity 3, one $M = 0$ state per copy), and $J = 0$ (multiplicity 2). A single computational basis state $|1100\rangle$ has nonzero overlap with *every* one of these. Build it as a TN of four single-qubit basis tensors.

```
In[56]:= ket1100TN = TensorNetwork[
  {{0, 1}, {0, 1}, {1, 0}, {1, 0}},
  {"q1"}, {"q2"}, {"q3"}, {"q4"}],
  {"q1", "q2", "q3", "q4"}];
```

Four disconnected single-qubit basis tensors.

```
In[57]:= ket1100TN["Hypergraph"]
```

```
Out[57]=
```



Contract.

```
In[58]:= ket1100 = TensorNetworkContract @ ket1100TN
```

```
Out[58]=
```

```
{{{{{{0, 0}}, {0, 0}}, {{0, 0}, {0, 0}}}, {{{0, 0}, {0, 0}}, {{1, 0}, {0, 0}}}}
```

Spin-2 probability via the unique SYT of {4}:

```
In[59]:= isoNormSq[ket1100, {YoungTableau[{{1, 2, 3, 4}}]]]
```

```
Out[59]=
```

$$\frac{1}{6}$$

Spin-1 probability via the three SYTs of {3, 1}:

```
In[60]:= isoNormSq[ket1100,
  {YoungTableau[{{1, 2, 3}, {4}}], YoungTableau[{{1, 2, 4}, {3}}],
  YoungTableau[{{1, 3, 4}, {2}}]]]
```

```
Out[60]=
```

$$\frac{1}{2}$$

Spin-0 probability via the two SYTs of {2, 2}:

```
In[61]:= isoNormSq[ket1100,
  {YoungTableau[{{1, 2}, {3, 4}}], YoungTableau[{{1, 3}, {2, 4}}]]]
```

```
Out[61]=
```

$$\frac{1}{3}$$

Physics readout: the three numbers are $1/6, 1/2, 1/3$, summing to 1. The $J = 2$ value matches the analytical $|\langle J = 2, M = 0 | 1100 \rangle|^2 = (1/\sqrt{6})^2 = 1/6$ (the unique $|J = 2, M = 0\rangle$ is the symmetric two-down state $\frac{1}{\sqrt{6}} \sum_{\sigma} |\sigma(1100)\rangle$). A single classical bitstring is a maximally non-eigenstate of $\vec{S}_{\text{tot}}^2$: it populates *every* total-spin sector available in its M -slice, with weights given by Young projection.

A 4-qubit valence-bond state: pure $J = 0$ dimer

Two bond singlets $|\Psi^-\rangle_{12} \otimes |\Psi^-\rangle_{34}$ form the simplest valence-bond crystal (the 4-site limit of the AKLT/RVB family on spin- $\frac{1}{2}$). It is a *pure* spin-0 state by construction, so Young projection onto every non- $\{2, 2\}$ shape must vanish.

The singlet $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ as a rank-2 tensor is $(i/\sqrt{2})\sigma_y$, so we get it from the built-in `PauliMatrix[2]` instead of hardcoding entries.

```
In[62]:= singletPair = (I/Sqrt[2]) PauliMatrix[2]
Out[62]=
```

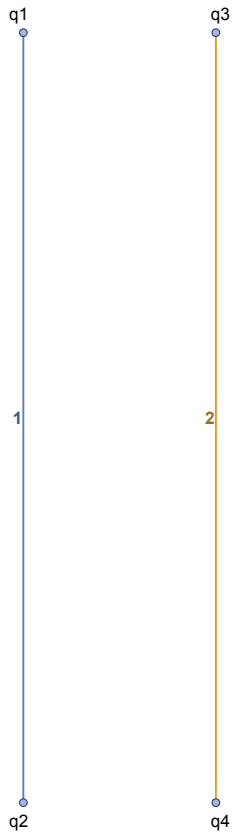
$$\left\{ \left\{ 0, \frac{1}{\sqrt{2}} \right\}, \left\{ -\frac{1}{\sqrt{2}}, 0 \right\} \right\}$$

Build the valence-bond state as a TN of two singlet tensors with no internal bonds.

```
In[63]:= vb4TN = TensorNetwork[
  {singletPair, singletPair},
  {{ "q1", "q2"}, {"q3", "q4"}},
  {"q1", "q2", "q3", "q4"}];
```

The hypergraph shows two disconnected rank-2 tensors, each carrying one pair of qubit legs.

```
In[64]:= vb4TN["Hypergraph"]
Out[64]=
```



Contract.

```
In[65]:= vb4 = TensorNetworkContract @ vb4TN
Out[65]=
```

$$\left\{ \left\{ \left\{ \{0, 0\}, \{0, 0\} \right\}, \left\{ \left\{ 0, \frac{1}{2} \right\}, \left\{ -\frac{1}{2}, 0 \right\} \right\} \right\}, \left\{ \left\{ \left\{ 0, -\frac{1}{2} \right\}, \left\{ \frac{1}{2}, 0 \right\} \right\}, \left\{ \{0, 0\}, \{0, 0\} \right\} \right\} \right\}$$

Spin-2 projection vanishes.

```
In[66]:= isoNormSq[vb4, {YoungTableau[{{1, 2, 3, 4}}]}]
Out[66]=
```

$$0$$

Spin-1 projection vanishes.

```
In[67]:= isoNormSq[vb4,
  {YoungTableau[{{1, 2, 3}, {4}}], YoungTableau[{{1, 2, 4}, {3}}],
  YoungTableau[{{1, 3, 4}, {2}}]}]
Out[67]=
```

$$0$$

Spin-0 projection captures the entire state.

```
In[68]:= isoNormSq[vb4,
  {YoungTableau[{{1, 2}, {3, 4}}], YoungTableau[{{1, 3}, {2, 4}}]}]
Out[68]=
1
```

Physics readout: a singlet-paired dimer state is in the kernel of $\vec{S}_{\text{tot}}^2$, occupying exactly the $J = 0$ block of dimension $\dim V_{\{2,2\}} \cdot \dim W_{\{2,2\}}(2) = 2 \cdot 1 = 2$. The two SYTs of $\{2, 2\}$ together span that 2D subspace, and the dimer $|\Psi^-\rangle_{12} \otimes |\Psi^-\rangle_{34}$ sits inside it. This is the calculation underlying valence-bond mean-field arguments for frustrated antiferromagnets: identifying which states a candidate ansatz can or cannot represent is exactly "which Young-tableau blocks does it have weight in".

Decoherence-free subspaces: Young projectors as collective-noise filters

The most direct quantum-information payoff of Young projection: *the multiplicity space V_λ is automatically invariant under collective noise*. If the environment couples symmetrically to all n qubits as a single-qubit unitary V applied identically to each ($V^{\otimes n}$), the action factors as $\text{id}_{V_\lambda} \otimes \rho_\lambda(V)$ on each Schur block, so collective unitaries act as the *identity* on V_λ and only mix vectors inside W_λ . Any state encoded in V_λ is therefore invisible to the noise: a *decoherence-free subspace* (DFS, Knill-Laflamme-Viola, Lidar-Whaley), no syndrome extraction needed. For 4 qubits, the $\lambda = \{2, 2\}$ block has multiplicity $\dim V_{\{2,2\}} = 2$, giving a 2-dimensional DFS: a noise-free *logical qubit* sitting inside the $J = 0$ sector. The point of using Young tableaux here is concrete: the packet identifies that protected subspace algebraically (no trial-and-error search), and `isoNormSq` lets you certify that an encoded state stays inside it after noise.

Apply a single-qubit unitary V identically on all n legs of a state tensor. In Einstein notation, $\psi'_{j_1 \dots j_n} = V_{j_1 i_1} \dots V_{j_n i_n} \psi_{i_1 \dots i_n}$: one copy of V per leg, contracted on the input index and exposing the output. The packet's `EinsteinSummation` reads exactly that, with integer labels (positive for inputs, negative for outputs); wrap in `Activate` to evaluate the resulting `TensorContract`.

```
In[69]:= applyCollective[V_, psi_] := With[{n = ArrayDepth[psi]},
  Activate @ EinsteinSummation[
    Append[Table[{-k, k}, {k, n}], Range[n]] → -Range[n],
    Append[ConstantArray[V, n], psi]]];
```

Draw a Haar-random 2×2 unitary in one line: `CircularUnitaryMatrixDistribution[n]` is the built-in Haar measure on $U(n)$, and a single `RandomVariate` gives one realization. The overall $U(1)$ phase makes no difference to the squared-norm tests below, so a Haar-uniform $U(2)$ is the same data as a Haar-uniform $SU(2)$ for this purpose.

```
In[70]:= SeedRandom[101];
V = RandomVariate[CircularUnitaryMatrixDistribution[2]]
```

```
Out[71]:=
{{0.641425 - 0.186374 i, 0.595824 + 0.445906 i},
 {0.523631 - 0.528819 i, -0.661883 - 0.0898433 i}}
```

Apply $V^{\otimes 4}$ to the valence-bond state from the previous section.

```
In[72]:= noisyVB4 = applyCollective[V, vb4];
```

Even though every amplitude of vb4 has changed, the entire weight still sits in the $\{2, 2\}$ block (the $J = 0$ sector). The DFS character of vb4 is preserved exactly.

```
In[73]:= N @ isoNormSq[noisyVB4,
 {YoungTableau[{{1, 2}, {3, 4}}], YoungTableau[{{1, 3}, {2, 4}}]]]
```

```
Out[73]:=
1.
```

No weight leaks into the spin-2 or spin-1 sectors.

```
In[74]:= N @ {isoNormSq[noisyVB4, {YoungTableau[{{1, 2, 3, 4}}]}],
 isoNormSq[noisyVB4,
 {YoungTableau[{{1, 2, 3}, {4}}], YoungTableau[{{1, 2, 4}, {3}}],
 YoungTableau[{{1, 3, 4}, {2}}]}]}
```

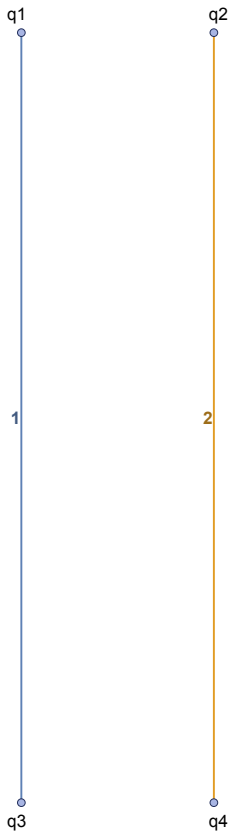
```
Out[74]:=
{0., 3.08149 × 10-33}
```

The DFS has dimension 2: a second linearly independent basis vector is the (13)(24) singlet pairing. Build it as a TN with the singlet tensors bonded across legs "q1" – "q3" and "q2" – "q4".

```
In[75]:= vb4bTN = TensorNetwork[
 {singletPair, singletPair},
 {"q1", "q3"}, {"q2", "q4"}},
 {"q1", "q2", "q3", "q4"}];
```

The hypergraph is again two disconnected rank-2 tensors, but now the singlet bonds straddle qubits (1, 3) and (2, 4) rather than (1, 2) and (3, 4).

```
In[76]:= vb4bTN["Hypergraph"]
Out[76]=
```



Contract.

```
In[77]:= vb4b = TensorNetworkContract @ vb4bTN;
```

It also lives in the $\{2, 2\}$ block, and stays there after collective noise.

```
In[78]:= N @ {isoNormSq[vb4b,
  {YoungTableau[{{1, 2}, {3, 4}}], YoungTableau[{{1, 3}, {2, 4}}]}],
  isoNormSq[applyCollective[V, vb4b],
  {YoungTableau[{{1, 2}, {3, 4}}], YoungTableau[{{1, 3}, {2, 4}}]}]}
Out[78]=
{1., 1.}
```

Physics readout: the 2D subspace spanned by $vb4$ and $vb4b$ (singlet pairings (12) (34) and (13) (24)) is the standard 4-qubit DFS used in trapped-ion qubit encoding (Kielpinski-Wineland) and NMR experiments. *Any* unitary that acts collectively on all four qubits fixes it pointwise (in the multiplicity-space sense), so encoding a logical qubit there gives complete protection against collective dephasing for free. The role of `YoungProject` is operational: it certifies that an encoded state has not leaked out of the DFS, without ever appealing to a noise channel.

Schur sampling: spectrum estimation from λ outcomes

If you have n identical copies of an unknown single-qubit state ρ and apply the Schur transform (Bacon-Chuang-Harrow 2006) followed by a measurement in the Young-diagram basis, the outcome is one

shape λ drawn with probability $\Pr(\lambda) = \text{Tr}(P_\lambda \rho^{\otimes n})$. As $n \rightarrow \infty$, the rescaled λ/n concentrates on the eigenvalue spectrum of ρ . This is the Keyl-Werner spectrum-estimation protocol (2001), provably sample-optimal. The point of using Young tableaux here is that the *Young diagram is the universal sufficient statistic* for the spectrum of ρ from independent and identically distributed samples: the entire information content of n identical experiments is in the random Young-diagram outcome λ , and computing $\Pr(\lambda)$ is just an isotypic norm.

Enumerate all SYTs of a given shape: all $n!$ permutations of $\{1, \dots, n\}$, split row-by-row into the shape via `TakeList`, kept only if `StandardTableauQ` passes. Brute force, but fine for $n \leq 8$ and trivially correct.

```
In[79]:= syts[lambda_List] := Select[
  YoungTableau[TakeList[#, lambda]] & /@ Permutations[Range[Total[lambda]]],
  StandardTableauQ];
```

The n -qubit computational basis as a list of rank- n tensors.

```
In[80]:= basisStates[n_] := Map[
  Function[idx, Normal @ SparseArray[idx -> 1, ConstantArray[2, n]]],
  Tuples[{1, 2}, n]];
```

For diagonal $\rho = \text{diag}(p, 1-p)$, the n -fold tensor $\rho^{\otimes n}$ is diagonal in the computational basis with eigenvalue $p^{\#0s} (1-p)^{\#1s}$ on each bit string, so $\Pr(\lambda) = \sum_i \|P_\lambda |i\rangle\|^2 \rho_{ii}^{\otimes n}$.

```
In[81]:= schurProb[shape_List, p_, n_] := Module[{tabs = syts[shape], weights},
  weights = (p^Count[#, 1] (1-p)^Count[#, 2]) & /@ Tuples[{1, 2}, n];
  Total @ MapThread[#2 isoNormSq[#1, tabs] &, {basisStates[n], weights}]];
```

Run it on $\rho = \text{diag}(0.7, 0.3)$ for $n = 4$ copies: probabilities of measuring the three available shapes.

```
In[82]:= N @ {schurProb[{4}, 0.7, 4], schurProb[{3, 1}, 0.7, 4], schurProb[{2, 2}, 0.7, 4]}
Out[82]:= {0.4141, 0.4977, 0.0882}
```

The three probabilities are approximately 0.41, 0.50, 0.09, summing to 1 (verified by the orthogonal-isotypic-projector property). They match the analytical Keyl-Werner formula

$\Pr(\lambda) = \dim V_\lambda \cdot s_\lambda(p, 1-p)$ with s_λ the Schur polynomial.

Repeat for $n = 6$ copies: now four shapes are available.

```
In[83]:= N @ Table[schurProb[lam, 0.7, 6],
  {lam, Select[IntegerPartitions[6], Length[#] <= 2 &]}]
Out[83]:= {0.205339, 0.434805, 0.313551, 0.046305}
```

Identify the most-likely shape (the "Schur estimate" of ρ 's spectrum) at each n , and show its rescaled λ/n converging to $\text{spec}(\rho) = (0.7, 0.3)$.

```
In[84]:= mostLikelyShape[p_, n_] := First @ MaximalBy[
  Select[IntegerPartitions[n], Length[#] ≤ 2 &],
  schurProb[#, p, n] &];

Table[With[{lam = mostLikelyShape[0.7, n]},
  {n, lam, N @ PadRight[lam, 2] / n}], {n, 4, 8, 2}]

Out[84]=
{{4, {3, 1}, {0.75, 0.25}},
 {6, {5, 1}, {0.833333, 0.166667}}, {8, {6, 2}, {0.75, 0.25}}}
```

Physics readout: at $n = 4$ the modal shape is $\{3, 1\}$ (rescaled $\{0.75, 0.25\}$); at $n = 6$ it shifts to $\{5, 1\}$ (rescaled $\{0.833, 0.167\}$); at $n = 8$ to $\{6, 2\}$ (rescaled $\{0.75, 0.25\}$). As n grows the rescaled mode converges to $(0.7, 0.3)$, exactly as Keyl-Werner predicts. Every quantum protocol that uses independent and identically distributed copies (quantum hypothesis testing, universal compression, entanglement concentration) reduces to *measuring this λ outcome*, and `YoungProject` is the constructive way to compute the probabilities.

Symmetric tensor networks: bond Hilbert spaces split by Young diagram

Tensor-network ansatzes that preserve $SU(2)$ (or $SU(d)$, $U(1)$, any compact group) have *bond* Hilbert spaces that automatically decompose into Young-diagram blocks. Each bond carries a label (λ, α) with λ a Young diagram and α a multiplicity index; the MPS/PEPS tensors are block-sparse with respect to this structure. The point of using Young tableaux for symmetric TNs is twofold: (i) the ansatz produces states with the *correct* quantum-number sector by construction (no contamination from other J values), and (ii) every contraction is *block-diagonal*: a $\chi \times \chi$ matrix-vector product is replaced by independent products on much smaller $\text{mult}_\lambda \times \text{mult}_\lambda$ blocks. The paclet's `TableauDimension` gives the per-sector multiplicities, so you can read off the *symmetric* bond dimension without building the MPS. Multiplicity of each spin- J in n qubits, returned as $\{J, \dim V_\lambda\}$ pairs.

```
In[85]:= spinSectors[n_] := Table[
  {(n - 2 k) / 2,
   TableauDimension[If[k == 0, {n}, {n - k, k}]]},
  {k, 0, Floor[n / 2]}];
```

For $n = 4$ qubits: one spin-2, three spin-1s, two spin-0s.

```
In[86]:= spinSectors[4]
Out[86]=
{{2, 1}, {1, 3}, {0, 2}}
```

For $n = 8$ qubits, the multiplicities grow: {14, 0, 28, 0, 20, 0, 7, 0, 1} when read out per integer $2J$ (here, fourteen spin-0 copies, twenty-eight spin-1, twenty spin-2, seven spin-3, one spin-4). The packet gets them in one line.

```
In[ ]:= spinSectors[8]
Out[ ]:=
{{4, 1}, {3, 7}, {2, 20}, {1, 28}, {0, 14}}
```

For an MPS half-cut at site $n/2$, the *non-symmetric* bond dimension upper bound is $2^{n/2}$. The *symmetric* bond dimension (used by SU(2)-preserving MPS like the symmetric DMRG implementations of Schollwöck, Singh-Pfeifer-Vidal, McCulloch, ITensor) is the sum of multiplicities of the left half over all J : each spin- J on the left can only pair with the matching spin- J on the right, so each multiplicity contributes one bond mode.

```
In[ ]:= bondCompare[n_] := With[{half = n/2},
  {n, 2^half, Total[spinSectors[half][[All, 2]],
  N @ 2^half / Total[spinSectors[half][[All, 2]]]};
Tabulate for n = 4, 6, 8, 10, 12 qubits.
In[ ]:= Grid[Prepend[bondCompare /@ Range[4, 12, 2],
  {"n", "raw bond 2^(n/2)", "symmetric bond", "ratio"}], Frame -> All]
```

```
Out[ ]:=
```

n	raw bond $2^{(n/2)}$	symmetric bond	ratio
4	4	2	2.
6	8	3	2.66667
8	16	6	2.66667
10	32	10	3.2
12	64	20	3.2

For the $n = 8$ half-cut concretely: each half is 4 qubits with multiplicities (1, 3, 2) for spin (2, 1, 0) from `spinSectors[4]`. The symmetric bond carries $1 + 3 + 2 = 6$ modes, organized as three independent blocks of sizes 1, 3, 2. A dense bond-matrix multiplication costs $6^3 = 216$ flops; the same operation done block by block costs $1^3 + 3^3 + 2^3 = 36$ flops, a $6\times$ saving from bookkeeping alone.

Physics readout: the savings ratio grows roughly like $\sqrt{n}$ (a known fact: the largest spin- J multiplicity in n qubits scales like $2^n / n^{3/2}$, so half-cut symmetric bond dim grows like $2^{n/2} / \sqrt{n/2}$). More importantly, the *correctness* aspect is free: a symmetric MPS targeting a Heisenberg ground state will return a state with definite total $J = 0$ by construction. Every entry in the `spinSectors` list is one Young-diagram-labeled bond mode the MPS has to track, and `TableauDimension` tells you how many degrees of freedom each carries before you build any tensor.

References

Decoherence-free subspaces:

- E. Knill, R. Laflamme, L. Viola, *Theory of Quantum Error Correction for General Noise*, Phys. Rev. Lett. **84**, 2525 (2000).
- D. A. Lidar, I. L. Chuang, K. B. Whaley, *Decoherence-free subspaces for quantum computation*, Phys. Rev. Lett. **81**, 2594 (1998).
- D. Kielpinski et al., *A Decoherence-Free Quantum Memory Using Trapped Ions*, Science **291**, 1013 (2001).

Schur transform and spectrum estimation:

- M. Keyl, R. F. Werner, *Estimating the spectrum of a density operator*, Phys. Rev. A **64**, 052311 (2001).
- D. Bacon, I. L. Chuang, A. W. Harrow, *Efficient Quantum Circuits for Schur and Clebsch-Gordan Transforms*, Phys. Rev. Lett. **97**, 170502 (2006).

Symmetric tensor networks:

- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, Annals of Physics **326**, 96 (2011).
- S. Singh, R. N. C. Pfeifer, G. Vidal, *Tensor network decompositions in the presence of a global symmetry*, Phys. Rev. A **82**, 050301 (2010).
- M. Fishman, S. R. White, E. M. Stoudenmire, *The ITensor Software Library for Tensor Network Calculations*, SciPost Phys. Codebases **4** (2022).

Young tableau background (textbooks):

- B. E. Sagan, *The Symmetric Group: Representations, Combinatorial Algorithms, and Symmetric Functions*, Springer GTM **203** (2001).
- W. Fulton, *Young Tableaux, with Applications to Representation Theory and Geometry*, Cambridge LMS Student Texts **35** (1997).
- M. Hamermesh, *Group Theory and Its Application to Physical Problems*, Dover (1989).